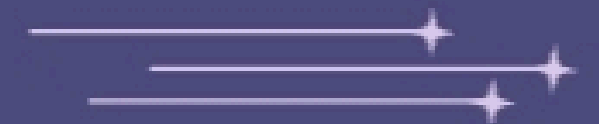


Example 2.2

Equal Complex Numbers

Finding Real Numbers x and y



Example 2.2

Find real numbers x, y if $(2+i)x+(1-i)y+2i-3$ and $x+(-1+2i)y+1+i$ are equal

Set up z_1 and z_2 :

$$\begin{aligned} z_1 &= (2+i)x + (1-i)y + 2i - 3 \\ &= (2x+y-3) + i(x-y+2) \end{aligned}$$

$$\begin{aligned} z_2 &= x + (-1+2i)y + 1 + i \\ &= (x-y+1) + i(2y+1) \end{aligned}$$

Given $z_1 = z_2$:

$$(2x+y-3)+i(x-y+2) = (x-y+1)+i(2y+1)$$

Equate Real & Imaginary Parts:

$$\begin{aligned} \text{Real: } 2x+y-3 &= x-y+1 \\ \Rightarrow \mathbf{x+2y} &= \mathbf{4} \end{aligned}$$

$$\begin{aligned} \text{Imag: } x-y+2 &= 2y+1 \\ \Rightarrow \mathbf{x-3y} &= \mathbf{-1} \end{aligned}$$

**Solving the two equations
simultaneously:**

$$\mathbf{x = 2, y = 1}$$